

Full Length Article

A many-body dissipative particle dynamics simulation of flow performance in capillary channel

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ABSTRACT

Fluid flow at the mesoscale has become an attractive topic of interest to improve the fluid transport behavior in reservoir formations. It is greatly advantageous to investigate the transport phenomena of small characteristic sizes by the many-body dissipative particle dynamics (MDPD) method. This work analyzes the wetting behavior between MDPD fluid particles and solid walls by tuning the attractive parameter, where an increase in attractive parameter value actually leads to weaker attraction in MDPD simulations. Parallel-plate flows for smooth and rough surfaces are used to investigate the transport property of MDPD fluid particles. The flow performance in capillary models is also studied. Results indicate that both surface and capillary contact angles increase with the rise of the attraction parameter. For smooth and rough walls, the velocity profiles show the parabolic shape. The velocities increase with increasing levels of attraction parameters and external forces. When the wettability of the solid wall decreases (or the attraction coefficient increases), the kinetic energy of MDPD fluid particles increases in comparison to the strongest wetting. The presence of a rough wall has a noticeable effect on the velocity distribution close to the wall, while the type of wall grooves has very little effect. Moreover, the different types of grooves do not affect the center velocity too much. The displacement particle number (the number of external particles entering the monitoring range) tends to rise with increasing external pressure and capillary diameter. Some conditions, such as low surface hydrophilicity, high external pressure, and large capillary diameter, can improve the flow performance. This MDPD simulation strategy provides a systematic study of flow transport behavior at the mesoscopic scale. It also can be readily extended to investigate more complex flow in formation conditions.

1. Introduction

Oil and gas resources, as the traditional energies, have a significant impact on economic development for many countries. High production is always desirable by oil companies and researchers. Hydraulic fracturing and waterflooding are the most commonly used methods for enhancing hydrocarbon production [1–4]. However, in low permeability reservoirs, such as shale gas, shale oil, and tight gas formations, injecting working fluid is difficult due to the high injection pressure [5,6]. Complexed fracture networks created by hydraulic fracturing improve the conductivity, reducing fluid friction. As a result, the fluid flow takes place in various channels. Besides flow in fractures, imbibition, back-flow, and displacement also occur in substrate capillaries during shut-in and exploitation [7]. Therefore, the investigation of flow

performance in fracture and capillary models is crucial to enhance production economically and efficiently.

The flow behavior is dependent on rock wettability, external forces, fracture geometries, and capillary characteristics. For instance, a typical strategy to enhance the flowability is to modify the rock surface wettability, which varies from hydrophilicity to hydrophobicity [8–10]. The surface wettability and flow performance can be characterized by Yong's contact angle [11–13] and flow experiments. Surface treatment in situ has been proven effective in achieving a desirable hydrophobic surface [14]. Chemical agents, such as nanoparticles, polymers, and surfactants, are widely employed as modifiers [15–19]. The flow characteristics are highly critical to estimating the resultant modification. Nowadays, most of the great efforts in this field are mainly made by diverse experiments. However, it is still challenging to investigate the

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transport performance in small-scale space by in situ experiments despite the rapid development of laboratory instrumentations for mesoscale confinement.

The powerful computation capacity of computers provides another possibility to understand the mesoscale behavior as a complementary method [20]. Mesoscale simulations have attracted extensive attention from scholars in numerous fields due to their low costs and high efficiency [21–23]. Additionally, a series of entangled influential elements have a remarkable effect on experimental results while the individual character still exhibits obscure [24]. Independent variation can be separately investigated and observed in the simulation process.

The dissipative particle dynamics (DPD) is a typical mesoscale particle method that was reported by Koelman and Hoogerbrugge in 1992 [25,26] and subsequently advanced by other scholars since that time [27,28]. The DPD method is based on Newton's second law of motion and uses the soft interaction forces. The soft potential allows larger length and time scales, reducing computation costs. The atomistic-scale details can be neglected, and the freedom degree of the total atoms decreases due to the coarse-graining treatment. This technique builds a link between macroscale and molecular scales [29,30]. It has many advantages over other methods such as molecular dynamics [31,32] and continuum computational fluid dynamics (CFD) [33,34]. However, the van der Waals loop is absent in the quadratic equation of state, which is not an accurate description of free surfaces. This limitation restricts the application of DPD. Many-body (or multibody) dissipative particle dynamics (MDPD) is a particle-based simulation method, which derives from the traditional DPD method [35,36]. Compared with the DPD approach, the conservation force consists of the short-range repulsive and long-range attractive interaction [37]. It is reliable to provide a fundamental understanding of the hydrodynamic behavior in complex fluid systems. MDPD method not only has relatively high computational efficiency but also can simulate the systems with multi-phase fluid flow phenomena [38,39].

Due to the distinctive advantages of simulating channel flow, it has been widely utilized to investigate flow performance. Xia et al. proposed a joint action of high-resolution FIB-SEM and MDPD models. This treatment provided a robust method to bridge gaps between micro/nano scales and macro scales, and it offered a potential way to study flow physics in nanoporous shales [30]. Liu et al. reported a novel induced flow through a bicontinuous nanoporous media using MDPD simulations. They found a dependence between steady-state fluid flow via the nanoporous models and pressure gradient, which was almost in coincidence with Darcy's law [37]. Chen et al. used MDPD simulations to investigate the spontaneous capillary imbibition and drainage, and they improved the Lucas-Washburn equation based on the simulated results [40]. Du et al. investigated flow performance and wetting on the rough and smooth surfaces for water droplets using MDPD simulation. The static and dynamic hydrophobicity were related to inherent hydrophobicity and surface roughness [41]. Zhou et al. studied the adsorption and aggregation performance of surfactants on rough and smooth surfaces by the MDPD method. Then, they analyzed the mechanisms of surfactant morphologies on the drag reduction effect [42].

Many research efforts have already been spent to characterize the flow behavior through flow experiments and DPD simulation methods. However, there is little information available on comprehensive investigations for flow performance in fracture and capillary models by many-body particle dynamic simulation. The main purpose of this work is a systemic investigation of flow performance based on the MDPD method. Benefiting from the advantages of MDPD, the surface and capillary wettability are investigated. The flow behaviors for smooth and rough surfaces under different conditions are analyzed. The effects of external forces and capillary diameters on displacement are discussed. It would be of great significance to widen the fundamental understanding and application of mesoscale flow.

2. Computational method

2.1. Many-body dissipative particle dynamics

MDPD is a coarse-grained method originated from DPD [43,44]. Although a detailed description of MDPD has been reported by Rao et al. [45] and Xia et al. [46], a more general introduction of MDPD is summarized here. The pairwise interaction forces between MDPD particles are decomposed into three parts, i.e., random force, dissipative force, and conservative force, as denoted by $\mathbf{F}_{ij}^R$, $\mathbf{F}_{ij}^D$, and $\mathbf{F}_{ij}^C$, respectively. These parts are given as expressed in Eqs. (1) to (3) [47–49]:

$$\mathbf{F}_{ij}^R = \sigma \omega_R(r_{ij}) \xi_{ij} \Delta t^{-0.5} \mathbf{e}_{ij} \quad (1)$$

$$\mathbf{F}_{ij}^D = -\gamma \omega_D(r_{ij}) (\mathbf{v}_{ij} \bullet \mathbf{e}_{ij}) \mathbf{e}_{ij} \quad (2)$$

$$\mathbf{F}_{ij}^C = A_{ij} \omega_c(r_{ij}) \mathbf{e}_{ij} + B_{ij} (\bar{\rho}_i + \bar{\rho}_j) \omega_d(r_{ij}) \mathbf{e}_{ij} \quad (3)$$

For the original DPD style, the random force coefficient σ is equal to $\sqrt{2k_B T \gamma}$, where T and k_B are the system temperature and Boltzmann constant, respectively.

γ is the dissipative parameter. ξ_{ij} is a random number with zero mean and unit variance. r_{ij} is the instantaneous distance between particles i and j . Δt refers to the timestep size. In these equations, $\mathbf{e}_{ij} = \frac{\mathbf{r}_{ij}}{r_{ij}} = \frac{\mathbf{r}_i - \mathbf{r}_j}{|\mathbf{r}_i - \mathbf{r}_j|}$ and $\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j$ denote the position unit vector and velocity vector difference between particles i and j , respectively. $\omega_R(r_{ij})$ and $\omega_D(r_{ij})$ are the distance-dependent weight functions, which can be denoted as $\omega_R(r_{ij}) = \sqrt{\omega_D(r_{ij})}$ [50]. The first term in Eq. (3) with $A_{ij} < 0$ describes the attractive interaction within a given cutoff radius of r_c , and the last term in Eq. (3) with $B_{ij} > 0$ is expressed as the density-dependent repulsive force within an interaction range of r_d . The weight functions in $\mathbf{F}_{ij}^C$ are given by $\omega_c(r_{ij}) = 1 - \frac{r_{ij}}{r_c}$ and $\omega_d(r_{ij}) = 1 - \frac{r_{ij}}{r_d}$, respectively. A common choice for local particle density $\bar{\rho}_i$ is $\bar{\rho}_i = \sum_{j \neq i} w_\rho(r_{ij})$, where $w_\rho(r_{ij})$ can be obtained as follows:

$$w_\rho(r_{ij}) = (1 - \frac{r_{ij}}{r_d})^2 \frac{15}{2\pi r_d^3} \quad (4)$$

For simplicity of discussion, dimensionless MDPD units are utilized in this simulation work. The modified velocity Verlet (MVV) algorithm, which was reported by Groot and Warren, integrates the equation of motion at every time step with the variable relaxation parameter $\lambda = 0.65$ [51]. The timestep interval is set as 0.01 MDPD time units. Sufficient steps are performed for data production. The weight functions are zero for interparticle distances larger than the cutoff distance (r_c or r_d). The computational domain is a rectangle box. The main parameters used in MDPD simulations are listed in Table 1. Gravity is not taken into account in this work since a small scale is set. The repulsive force coefficient B is set as 25. The attraction parameters between identical particles are kept at -40 . The density of fluid particles is about 6. The density of solid particles is approximately 11, which is denser than the fluid density to prevent the fluid particles from penetrating. The large-scale atomic/molecular massively parallel simulator (LAMMPS) [52] is selected for all particle dynamics studies.

Table 1
Parameters for present MDPD simulations.

Parameter	Symbol	MDPD unit
Mass	m	1.0
Dissipative intensity	γ	18.0
Cutoff distance (attr.)	r_c	1.0
Cutoff distance (rep.)	r_d	0.75

2.2. Contact angle

Assuming the MDPD fluid is water, the hydrophobic surface has a marked impact on flow performance, and contact angles (θ) can characterize hydrophobicity. Thus, the determination of contact angles between fluid particles and the solid substrate is an important work. Initially, the three-dimensional droplet particles are placed on the center of the substrate, as displayed in Fig. 1. The MDPD fluid particles aggregate together and present a spherical appearance in the three-dimensional view. As pointed out previously by Du et al. [41], their method is used to determine contact angles. Periodic boundary conditions are set in both x and z directions. The frozen particles are used as boundary walls due to a significant computation efficiency [53]. Fig. 2 shows the contact angles in half-sealing (sealing at one end) capillary. A half-sealing capillary model consists of fluid particles and capillary wall particles. According to the method proposed by Chen et al. [40], the contact angles between the meniscus and the wall are evaluated. The attractive parameter between the left wall (the green particles in Fig. 2) and fluid particles is set as -40 . The MDPD method evaluates wetting properties between the solid wall and fluid particles under different attractive parameters (A_{12}).

2.3. Parallel plates flow

The geometric characteristic of the fracture network created by hydraulic fracturing is highly complex. The transportation in fractures is simplified as the flow between two planar plates (or Poiseuille flow [54,55]). As shown in Fig. 3, the transport properties of smooth and rough walls (or groove walls) are investigated. The periodic boundaries are set in the x and z directions. The solid walls consist of MDPD particles, and the wall particles are restrained in the pre-equilibrated positions by the spring force

$$F_s = k_s(r_i(t) - r_i(0)) \quad (5)$$

with the $k_s = 12.5$. This interaction results in the solidification of the wall boundary. Compared with rigid walls, this method produces a desirable solid-liquid interface with weak density and temperature oscillations. A rational k_s makes the net density profile smooth and renders the smaller width of the interfacial region [56]. Simulation results show that applying spring force to the solid particles has a scant effect on the

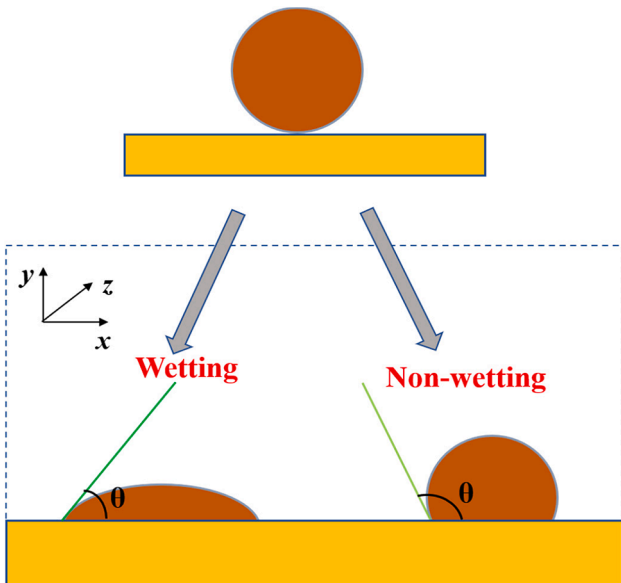


Fig. 1. Schematic representation of the surface contact angles for solid substrate.

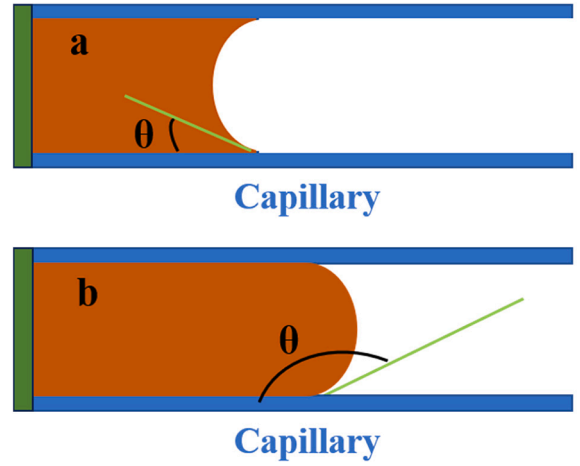


Fig. 2. Schematic representation of the contact angles for half-sealing capillary with MDPD fluid particles: (a) hydrophilicity. (b) hydrophobicity.

permeation of fluid particles. Various fluid-wall interactions are achieved by tuning the long-range attractive coefficient (A_{ij}). The body force works on fluid particles to make the fluid flow along the x direction between two plates, which results in a velocity field [57]. The simulation box is divided into narrow bins with $\Delta y = 0.02$ to achieve the velocity profiles. We specify the thickness of spatial bins in the x and y dimensions to be 0.1 for velocity color distribution, respectively.

2.4. Capillary displacement

To characterize flow performance in the capillary with external force, a few simulations are carried out. In the field of simulation box size, it is set to be 105 in the x direction. The different diameters of capillaries are chosen in different simulation cases. The diagrammatic sketch and snapshots of MDPD particles are displayed in Fig. 4 and Fig. 5, respectively. The initial state of the present model consists of four components, i.e., capillary, baffle, and two kinds of fluids (denoted as FL and FR, respectively), as shown in Fig. 4. The cylinder capillary with a rectangular outer shape is modeled. Two different wettability fluids are filled at the left and right sides of the capillary, respectively. These fluids are initially in contact with each other. A rigid baffle attaches to the left fluid (FL) and can freely move along the x -direction. The external force is applied to the baffle, and then fluids are forced to flow. Each capillary wall particle is applied a spring force to tether it to its initial location, as mentioned in Section 2.3. This initial model ($t = 0$ MDPD time units) is shown in Fig. 5(a). As time advances ($t = 100$ MDPD time units), the fluid particles are forced to move under external force, as shown in Fig. 5 (b).

The effects of attractive coefficients and cylinder diameters on displacement behavior are investigated. The attraction coefficients, i.e., FL-wall, FL-FR, and FR-wall, are denoted as $A_{FL-wall}$, A_{FL-FR} , and $A_{FR-wall}$, respectively. The FR particles are forced to flow by FL. The number of displacement particles in a system slice with a width of 1 MDPD unit is monitored over time, as shown in Fig. 4. The coordinate origin is set at the FL-FR interface. When inappropriate parameters are chosen, the particle oscillation occurs close to the interface. Here, we choose $x = 3$ MDPD units along the x direction as the origin of the monitoring range to decrease this effect. Once the external particles enter this range, they are counted. The displacement particle number at $t = 0$ MDPD time units stands for the original particle number over the monitoring region before the forced flow. After the flow begins, the displacement particle number accumulates with time.

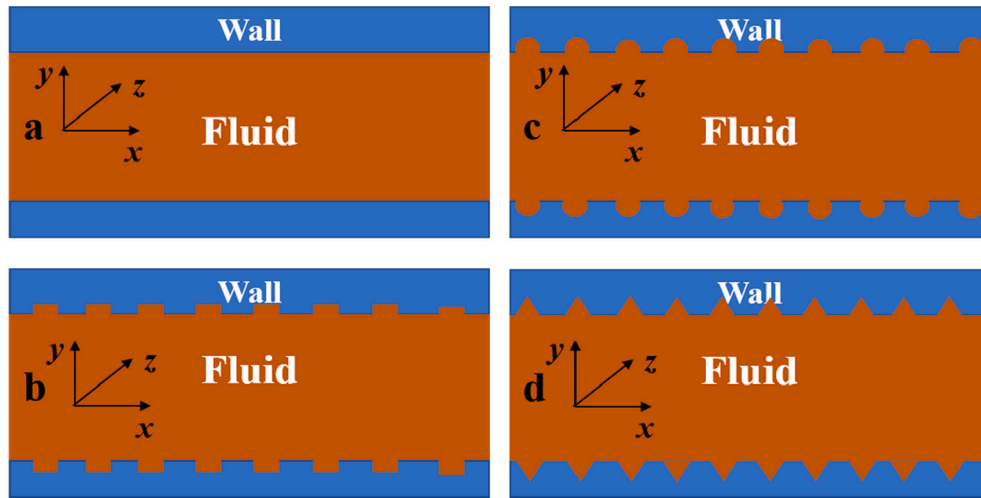


Fig. 3. Schematic representation of flow channels: (a) smooth wall. (b) rectangle groove wall. (c) cylinder groove wall. (d) triangle groove wall.

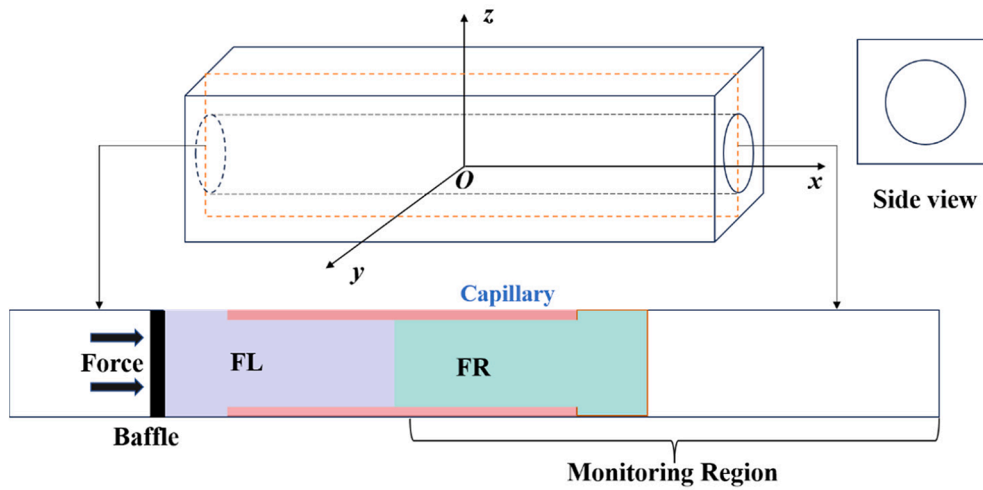


Fig. 4. Schematic illustration of the components of capillary displacement.

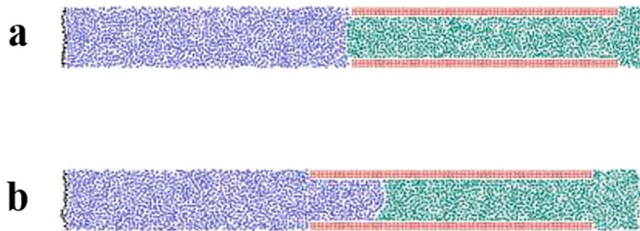


Fig. 5. The snapshots of the slice view of the system at (a) $t = 0$ MDPD time units and (b) $t = 100$ MDPD time units.

3. Results and discussion

3.1. Contact angle analysis

The dynamic process of fluid is well correlated with the wetting property. Generally, when the contact angle between the fluid and wall is more than 90° , the fluid is considered to be a nonwetting phase (assuming the MDPD fluid is water); otherwise, the fluid is regarded as a wetting phase. If the contact angle is 0° , it is known as total wetting behavior. The contact angles in the range from 0° to 90° usually correspond to the partially wetting behavior [58].

The contact angles are closely related to the attraction parameter. Firstly, the wetting behavior between the droplet and substrate wall is investigated under a series of attractive coefficients. The snapshots of

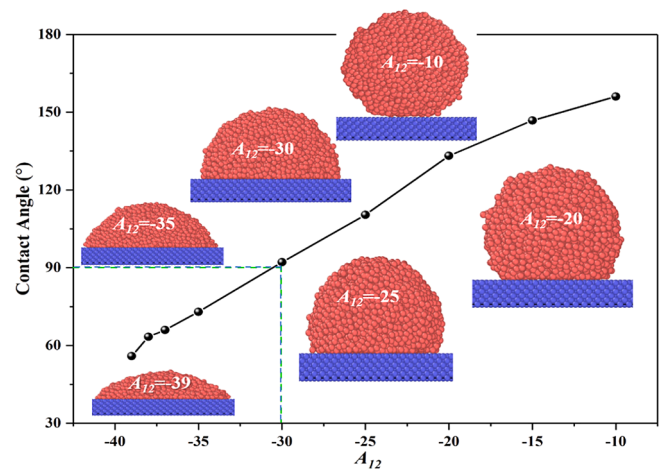


Fig. 6. Two-dimensional snapshots of surface contact angles with different wall-particle attractive coefficients.

droplet morphologies in Fig. 6 present the change of contact angles with an increase in the attractive coefficient. The surface contact angles monotonically increase with the rise of the fluid-wall attractive parameter (A_{12}). The contact angles change from $\sim 55^\circ$ to $\sim 156^\circ$ as attractive parameter ranges from -39 to -10 . When the attractive parameter is less than -30 , many fluid particles tend to stick to the solid surface, i.e., they present the wetting behavior. Once the attractive coefficient is beyond -30 , the attached area between the wall and the droplet gradually reduces, indicating that the nonwetting property of the surface rises.

There are various types of capillaries, such as total-sealing, half-sealing, and total open capillaries, in the formation. For simplicity, the contact angles of the half-sealing capillary are studied. At the capillary channel, the contact angles are different from those in Fig. 6. But the tendency of half-sealing capillary contact angles is in well accordance with the surface wetting angles. As evidenced in Fig. 7, it is observed that the capillary contact angles increase with increasing fluid-wall attractive coefficient (A_{12}) ranging from -40 to -10 , which varies from hydrophilicity to hydrophobicity. The shapes in the inset of Fig. 7 present the inward menisci when the attraction parameter is less than -30 . The inward meniscus disappears with a decrease in the hydrophilicity of the wall, and then the outward meniscus gradually occurs with increasing hydrophobicity of the wall. The meniscus with an attraction coefficient of -40 shows the maximum inward curvature. The meniscus with an attraction parameter of -10 has the maximum outward curvature.

3.2. Flow characteristics of parallel plates

The Poiseuille flow is usually of interest to scholars since it is significant in numerous practical applications. The analyses of the evolution and dynamics of flow behavior will deepen the understanding of transport phenomenon. In the current work, the flow performance can be obtained by varying the fluid-wall attraction coefficient (A_{13}) and surface roughness, and the thus-obtained results are exhibited in Figs. 8–12.

With the attraction parameter rising from -40 to -5 , the parabolic velocity profile changes accordingly, as shown in Fig. 8. On the one hand, in the center region, the apparent velocity increase can be observed with the continuous increase of the attraction coefficient. On the other hand, the velocity in the near-wall region is lower than that in the center region. The velocity profiles exhibit minor differences when the attraction coefficients are set to -40 and -30 . It indicates that the velocity distribution for smooth hydrophilicity surfaces is less

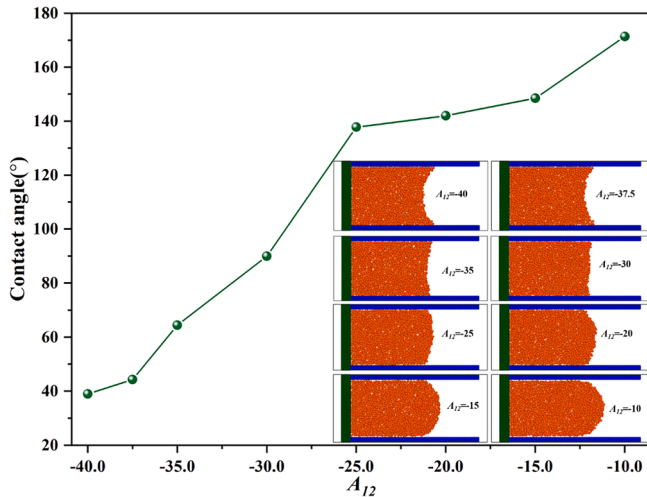


Fig. 7. Half-sealing capillary contact angles under different attraction coefficients.

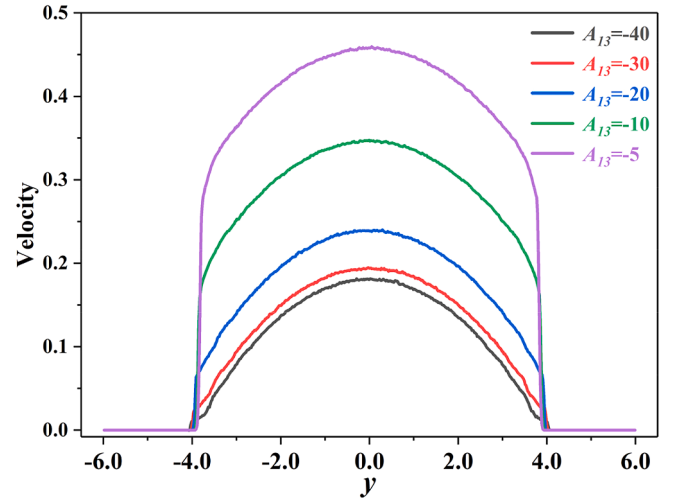


Fig. 8. Velocity profiles along the y direction for the smooth solid wall at different attraction parameters with $F = 0.025$.

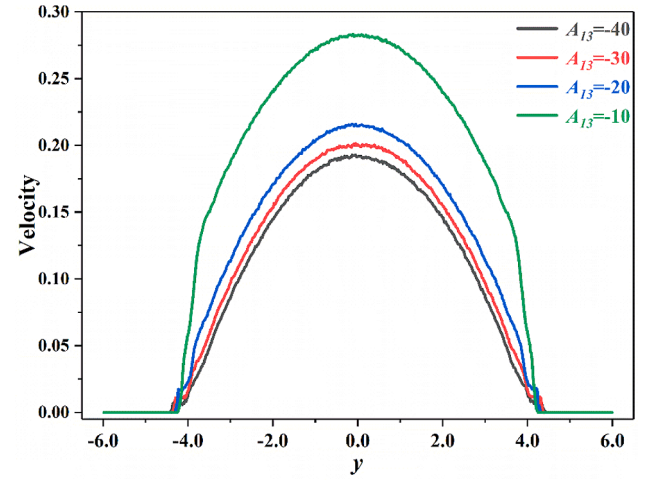


Fig. 9. Velocity profiles of the rough solid wall (rectangle-shaped grooves) with $F = 0.025$.

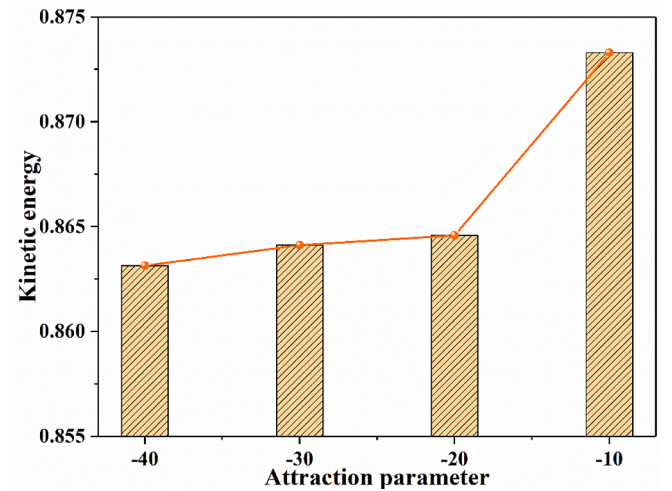


Fig. 10. Kinetic energy of rough-surface (rectangle-shaped grooves) flow under different attraction parameters with $F = 0.025$.

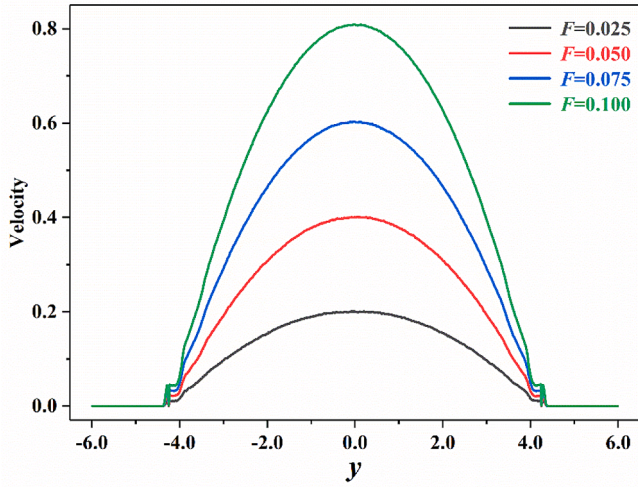


Fig. 11. Change in the velocity upon varying external forces for rough surfaces (rectangle-shaped grooves) for $A_{13} = -30$.

susceptible to the attractive force within this range. In addition, the slip velocities and the curvatures of velocity curves progressively increase with increasing attractive coefficients of fluid–solid. Therefore, the data reported here reveals that more fluid particles pass the channel in a hydrophobic condition than in a hydrophilic condition.

Fig. 9 shows the effect of the attraction parameter on the velocity profile for a rectangle groove surface. The velocity profiles exhibit a similar tendency to the smooth surface in Fig. 8. However, the remarkable difference occurs near the wall boundary. On the one side, the velocity slip behavior weakens, even though the attraction parameter is -10 . The presence of wall grooves restricts velocity slip close to the wall. On the other side, the wall influences on the velocity distribution over a wide range. High flow in transportation usually means low friction losses under the same condition, thereby decreasing energy

consumption. The kinetic energy of fluid particles is displayed in Fig. 10. The kinetic energy is observed to increase with increasing attraction parameters. As the attractive parameter between solid wall and fluid particles varies from -40 to -20 , the kinetic energy slightly increases. Wall grooves partly counterbalance the enhancement effect from the attraction parameter. A remarkable rise in kinetic energy can be observed when the attraction parameter is set as -10 , indicating an insignificant effect of wall grooves on flow behavior. Similar kinetic energy and velocity tendencies are present, signifying that it is an effective method to enhance the flow by improving surface hydrophobicity.

The effect of external force (F) on the velocity profile is analogous to that of the attraction parameter, as displayed in Fig. 11. Firstly, the velocity tends to rise with increasing external force. In addition, the parabolic-shaped velocity profile shows a successively increasing curvature, especially in the center region. Another observation from Fig. 11 is that the slip velocity rises slightly near the wall boundary. The solid wall still plays a leading role in determining velocity, showing the minimum difference in velocity. Thus, enhancing the flowability by increasing the external force is effective.

The real fracture surface has grooves or crevices, and the roughness has an evident impact on flow performance, particularly close to the wall. Three types of grooves are fabricated on solid substrates to evaluate the impacts of groove structures. Fig. 12 presents the velocity distribution in the flow channel. Colors are utilized to express the velocity magnitude (v_x). The velocity distribution shows apparent laminar-shape flow behavior for the smooth surface (as shown in Fig. 12 (a)). The velocity gradually increases, ranging from the wall boundary to the center region. For the groove surfaces (as shown in Fig. 12 (b), (c), and (d)), the surfaces have an impressive effect on the velocity distributions, especially at the solid wall. Although rectangle-shaped grooves are fabricated, the velocity distribution displays the sawtooth fluctuation. This influence slowly vanishes as it approaches the center region. The similar velocity distributions are also noticed in Fig. 12 (c) and (d). The velocity of cylinder-shaped grooves goes deeper into the wall in

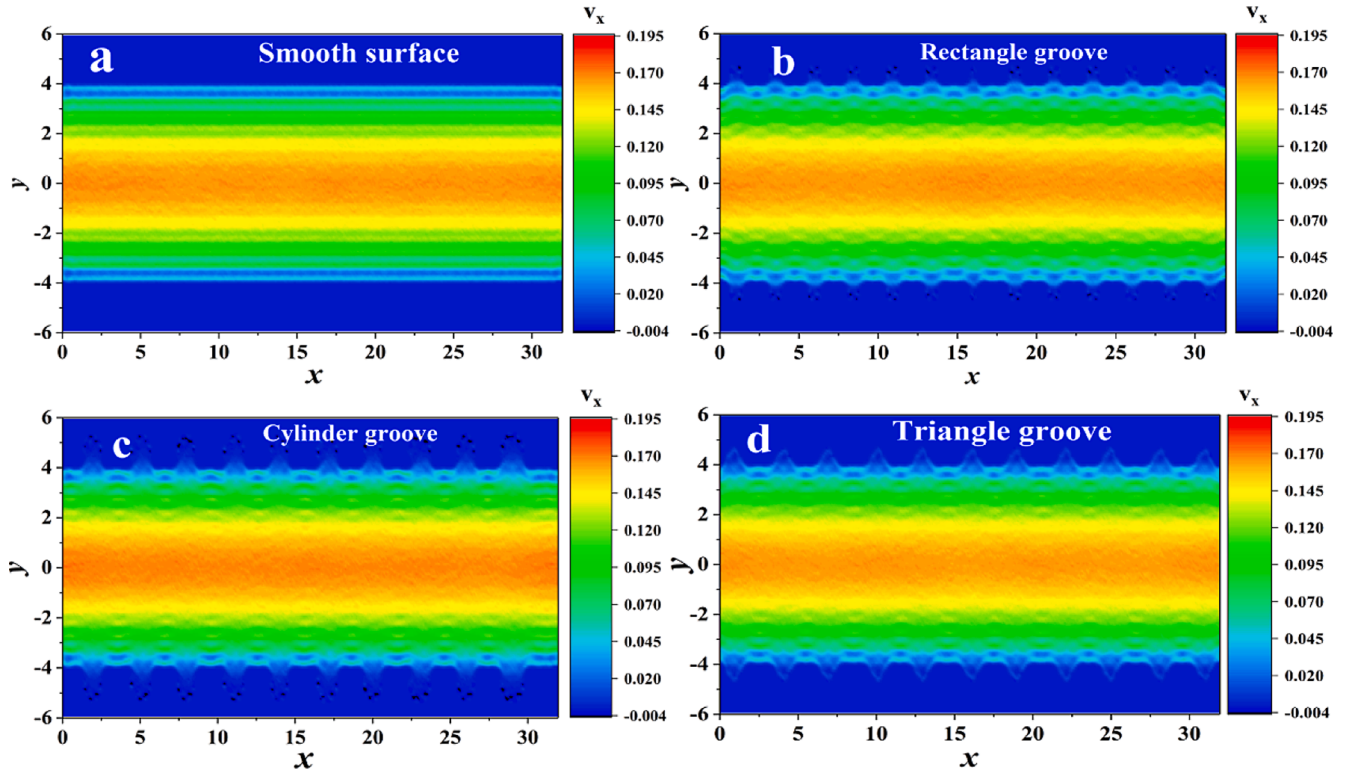


Fig. 12. Velocity distributions with different wall grooves for $A_{13} = -30$: (a) smooth surface, (b) rectangle grooves, (c) cylinder grooves, and (d) triangle grooves.

comparison to the rectangle-shaped grooves. The velocity of triangle-shaped grooves exhibits a sawtooth distribution, which is similar to the rectangle-shaped grooves. As mentioned above, conclusions can be reached. On the one hand, the appearance of grooves has a pronounced effect on the velocity color distribution close to the wall. On the other hand, the type of wall grooves has very little effect on velocity distribution. Moreover, the kinds of grooves do not affect the center velocity distribution too much.

3.3. Capillary displacement analysis

Fluid-fluid displacement is one of the most widely used methods for flow transport in geological mesoporous. The forced displacement process can be used to evaluate fluid mobility. Diverse factors, such as external pressure (f), fluid-wall interaction, and capillary diameter (d), are closely related to the performance of fluid-fluid displacement [59].

The external pressure in practical application is a crucial parameter, indicating flowability in the capillary to some degree. Low external pressure is always desirable due to energy savings. Fig. 13 displays the curves obtained under different external pressures (fixing $A_{FL-wall} = -10$, $A_{FL-FR} = -20$, $A_{FR-wall} = -40$, $d = 6$, while varying external pressures). All curves present slight fluctuation with time. It is worth noting that the displacement particle number is susceptible to external pressure. When external pressure is not more than 8.25, the displacement particle number is a negligible dependence on time, demonstrating that external force is not enough to overcome the adhesion force. When external pressure is set as 16.50, external pressure can make the fluid flow, and the displacement particle number is positively related to the time. The displacement particle number rises with the continuous increase of time and external pressure. It is worth mentioning that the slopes of curves increase with increasing external pressures, which means that it is highly prone to be displaced under high external pressure.

The diameter has an obvious effect on transport phenomena at micro/nano-scope channels. The displacement performance under different diameters is investigated, as shown in Fig. 14 (fixing $A_{FL-wall} = -10$, $A_{FL-FR} = -20$, $A_{FR-wall} = -40$, $f = 33.00$, while varying diameters). It is worth noting that the displacement particle number for all curves maintains the linear relation over the displacement time. On the one hand, the displacement particle number increases when the diameter rises from 5 to 8 MDPD length units. On the other hand, the level of slopes of displacement curves is also observed to increase with increasing diameter, indicating that the larger the diameter of flow results in more particles and less flow time under the same condition. It is reasonably deduced that the small diameter channel can reduce the fluidity of fluid and increase the displacement cost.

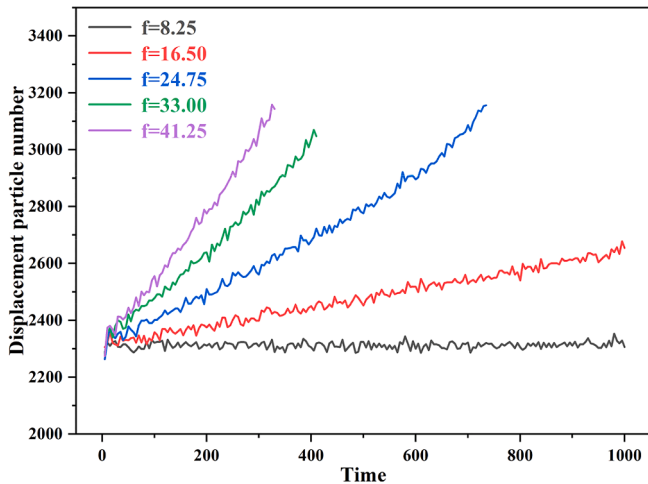


Fig. 13. Change in displacement particle number with time under different external pressures.

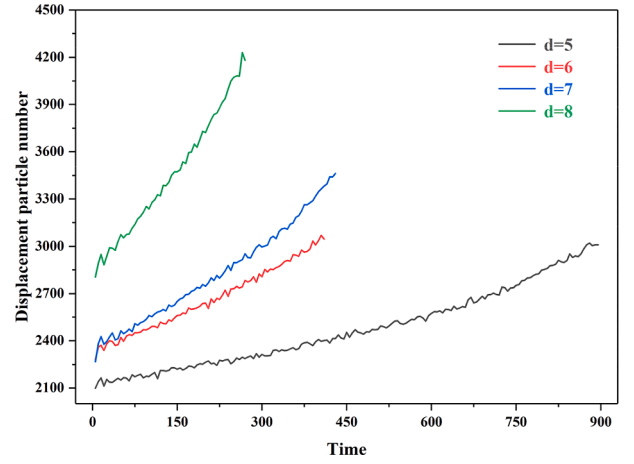


Fig. 14. The time-dependent displacement particle number for different diameters.

Here, the effect of the FL-wall attraction parameter ($A_{FL-wall}$) on the displacement performance is studied. The thus-obtained results are depicted in Fig. 15 (fixing $d = 6$, $A_{FL-FR} = -20$, $A_{FR-wall} = -40$, $f = 33.00$, while varying $A_{FL-wall}$ among -5 , -10 , -20). For all cases, there is a linear dependence of displacement particle number with increasing time. An increase in the attractive force parameter from -20 to -5 exhibits no remarkable change in displacement particle number with time. The simulation result seems discrepant with a few actualities obtained by experiments [8]. Besides the alternation of the attractive parameter between FL fluid and solid wall, other interface properties are affected accordingly, such as the attraction parameter between FL and FR (or miscibility). For instance, FL denotes surfactants instead of original water. However, in this simulation work, the attraction parameter between FR and FL remains constant, which differs from the experimental conditions. It is not an effective way to evaluate the displacement performance only by changing the attractive parameter between FL and solid wall. The experiments usually present comprehensive results, including many influential factors. Our subsequent work is to study reasonable parameters by combining simulations and experiments.

3.4. Discussion of the influencing factors on fluid-wall interaction

The results above present the fluid performance in fracture and capillary models. This section analyzes the possible physical factors or experimental conditions that could lead to stronger fluid-wall

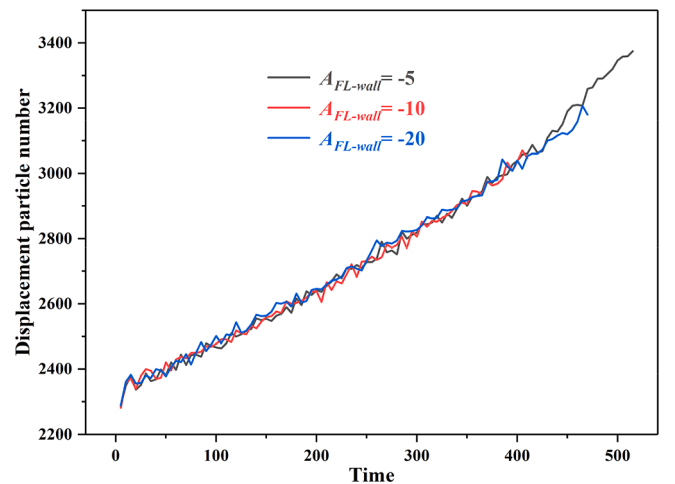


Fig. 15. Displacement particle number as a function of time for different attraction coefficients.

interactions to bring more insight into further study. Wettability is the critical factor in evaluating fluid-wall interaction in reservoir formations, which affects the fluidity of fluids. For instance, Zhang et al. pointed out that the rock-wetting characteristics controlled the capillary forces, which dominated the degree of spontaneous water imbibition [60]. Hence, the exact determination of wetting characteristics of a wall/fluid system is of utmost significance. There are numerous causes that affect wettability in reservoir formations.

On the one hand, we take shale reservoirs as an example [61]. Shale primarily contains clay minerals and brittle minerals. The hydrophilicity for lipophilicity is weaker than that of four clay minerals (montmorillonite, chlorite, kaolinite, and illite). The temperature affects the water concentration of clay mineral surfaces and thereby has a certain effect on the wettability. These results are similar to the experimental results [62,63]. Small pressure changes have an insignificant effect on wettability [64,65]. The lower the amount of NaHCO_3 added to water, the stronger the wettability of clay mineral surfaces by water.

On the other hand, we discuss the wetting characteristics of carbonate rocks [66]. The contact angles decrease slightly with decreasing pressure, indicating that surfaces turn slightly more hydrophilic. The contact angles decrease with increasing operating temperature for pure calcite samples. However, mixed tendencies are observed for dolomitic samples. Experimental results show that dolomitic-based and calcite-dominant samples were relatively more hydrophobic and hydrophilic, respectively. The surfaces with relatively higher roughness display much higher contact angles.

Additionally, the wettability (or contact angle) varies with the changes in temperature, pressure, water salinity, elemental composition, and surface roughness during the production of oil and gas. Hence, the wettability or fluid-wall interaction is quite complex.

4. Conclusions

As a coarse particle method, MDPD establishes a connection between atomistic and macroscopic models, and it shows great potential in a wide range of engineering applications. MDPD is quite reliable for investigating the fluid flow in mesoscale channels. The droplet and capillary contact angles are well visualized, and then the contact angles are determined. The wettable behavior between the solid substrate and fluid particles can be controlled by tuning the attractive coefficients of the conservative force term. The simulation results illustrate that the contact angles increase with the increase of the attraction parameters. If the MDPD fluid is assumed to be water, the hydrophilicity decreases while hydrophobicity rises. The effects of attraction coefficient, external force, and surface property on the velocity profiles are investigated. The velocities tend to rise with increasing attraction parameters at the same position for smooth and rough surfaces. Similarly, the velocity profiles have strong relations with the external forces. Compared with the kinetic energy with an attraction coefficient of -40 , other attraction parameters show the high kinetic energies in transportation, indicating that it is an effective method to increase flowability by enhancing hydrophobicity. The presence of grooves has an obvious effect on velocity distribution approaching the wall, and the types of grooves have little effect on the center velocity distribution. The flow properties in capillary models are also discussed. There is a threshold value of external pressure, only above which left fluid particles can enter the right capillary. With an increase in the external pressure, the displacement particle number increases at the same condition, revealing that large external pressure improves flowability. The larger the diameter of the capillary, the more MDPD fluid particles they are and, consequently, more accessible to displace. The next step of this study is to investigate how coupled factors affect displacement. The MDPD strategy is a computationally efficient method to conduct flow simulations. It also provides a potential method for the flow behavior in fracture and capillary models, which is quite significant for exploiting hydrocarbon.

CRediT authorship contribution statement

Liang Xing: Writing – review & editing, Validation, Supervision, Software, Investigation, Formal analysis, Data curation. **YanJun Zhang:** Writing – original draft, Resources, Methodology, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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